

Design Verification Plan and Report (virtual-test version) — final design V8**Document no.:** VBF-T3R1NOFORCE-DVPR-01 · Case `exp/t3\_r1\_noforce` · 2026-08-25

All result values mechanically taken from simulation output files; uncovered conditions honestly written "N/A".

#	Verification item	Condition	Result value	Determination (vs entry-0 criteria)	Source
1	1C discharge capacity	1C discharge, 298.15 K ambient, DFN + lumped thermal	2.6683 Ah	PASS (≥ 2.0 Ah)	cell/r4_v8_1c_dfn.json:capacity_ah
2	5C discharge capacity retention	5C discharge, 298.15 K ambient, DFN	2.6261 Ah; retention 0.9842	PASS (≥ 0.95)	cell/r4_v8_5c_dfn.json:capacity_ah ÷ cell/r4_v8_1c_dfn.json:capacity_ah (cell/r4_v8_derived.json)
3	Power density	contract formula $V_{OC}^2/(4 \cdot DCR)/\text{mass}$ on 1C discharge	73121.8 W/kg	PASS (≥ 4000 W/kg)	cell/r4_v8_energy.json:power_density_w_kg
4	4C fast-charge temperature rise	4C charge at 45 °C ambient, DFN + lumped thermal	T_max 323.57 K (50.4 °C)	PASS (≤ 333.15 K)	cell/r4_v8_4c45_dfn.json:T_max_K
5	4C fast-charge plating	same as #4 + plating module; negative surface potential < 0 V → plated	anode potential min +0.0119 V → plated = false	PASS (plated = false)	cell/r4_v8_4c45_dfn.json:anode_potential_v
6	Voltage window	parameter set cut-offs	2.5 – 4.2 V	PASS (upper bound never exceeded in protocols)	base parameter set
7	Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
8	Overcharge to thermal runaway	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
9	Crush / drop	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
10	Cycle life	—	N/A (requires aging model; not simulated in this case)	N/A	—
11	Rate-pulse internal resistance	—	N/A (requires physical experiment)	N/A	—

Conclusion

All 6 simulation-verifiable items PASS against the entry-0 criteria (mechanical `bda log-evaluate` evidence, round 4). Uncovered items (#7–#11) are beyond the pure-simulation boundary and are cited directly as paper limitations. Verification notes: the C-rate protocols run at the cell's own rated C-rates (Nominal cell capacity override, probe-verified cell/probe_nominal_1c_dfn.json); the plating margin is +11.9 mV — thin but positive, flagged in the DFMEA for production-tolerance control.